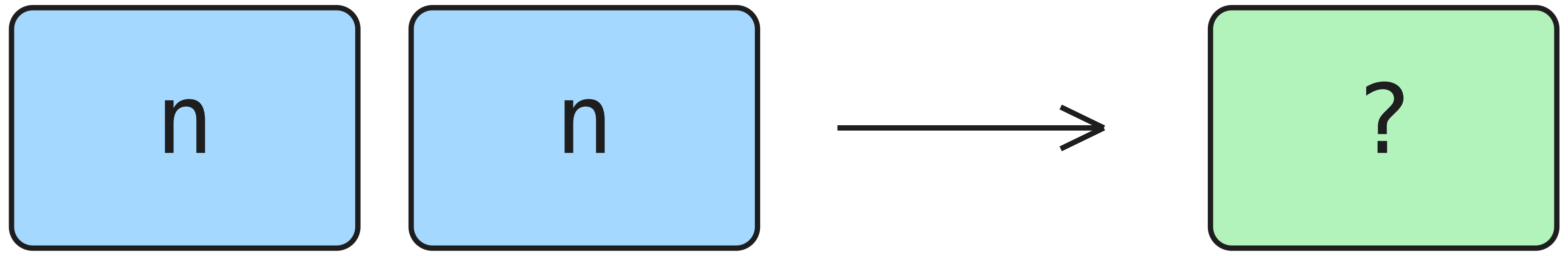
# One more character of memory.

The model still predicts one next token.

## BIGRAM


1 context + 1 target

## TRIGRAM



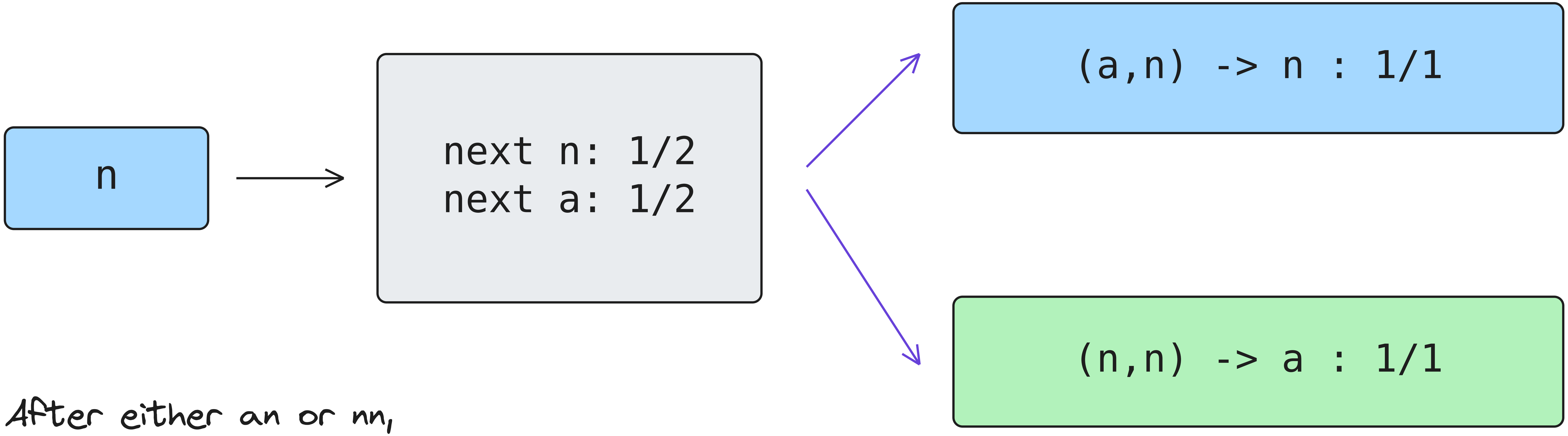

2 context + 1 target

Training names stay the same: anna ava

Trigram counts three tokens in the window, not three tokens to predict.

# Split one row into two questions.

Training examples: anna and ava. Counts shown before smoothing.

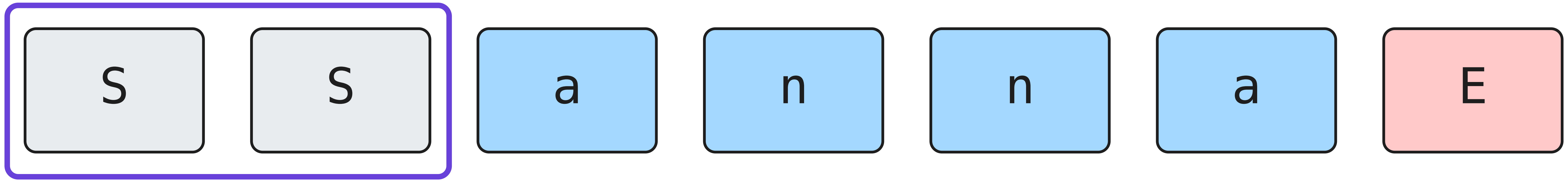



After either an or nn,
the bigram uses this row.

More specific questions. Fewer observations behind each answer.

# Two START pads. One END target.

Pads supply context; they are never predicted.




initial context

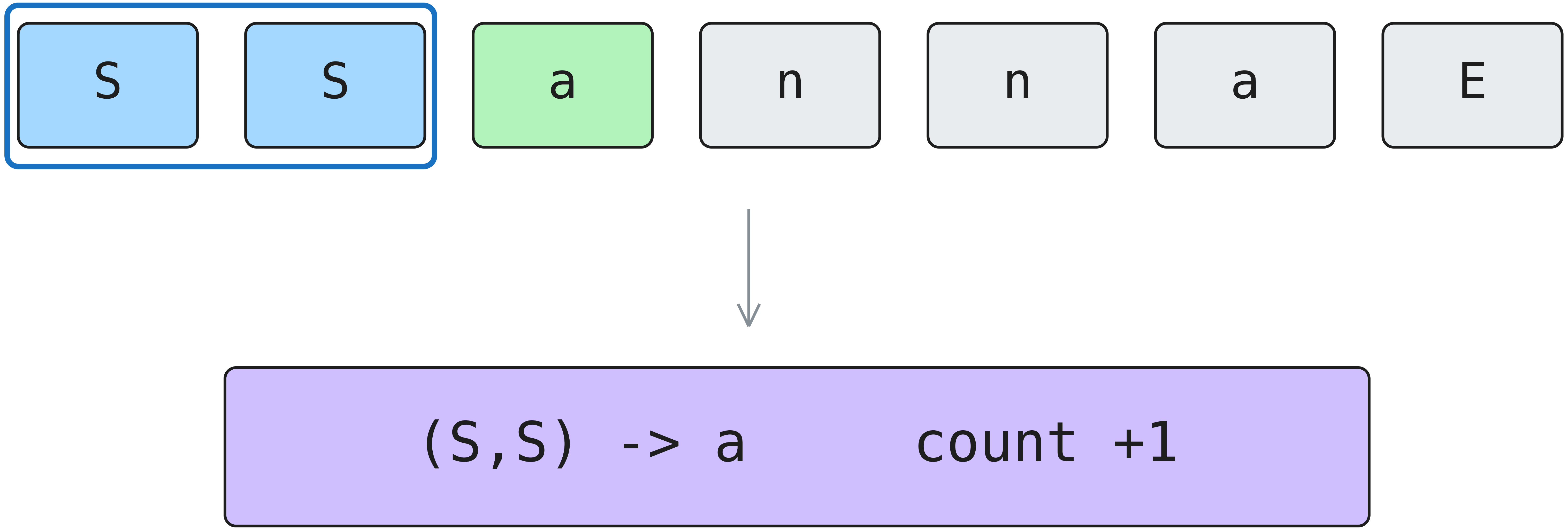
four characters + one END

Reset to (S,S) at the start of every name.

anna still contributes 5 predictions; ava contributes 4. Total = 9.

# Slide the window: anna, step 1 of 5.

Blue cards are the context. The green card is the observed target.




Move forward one position. Keep the last two tokens as context.

# Slide the window: anna, step 2 of 5.

Blue cards are the context. The green card is the observed target.

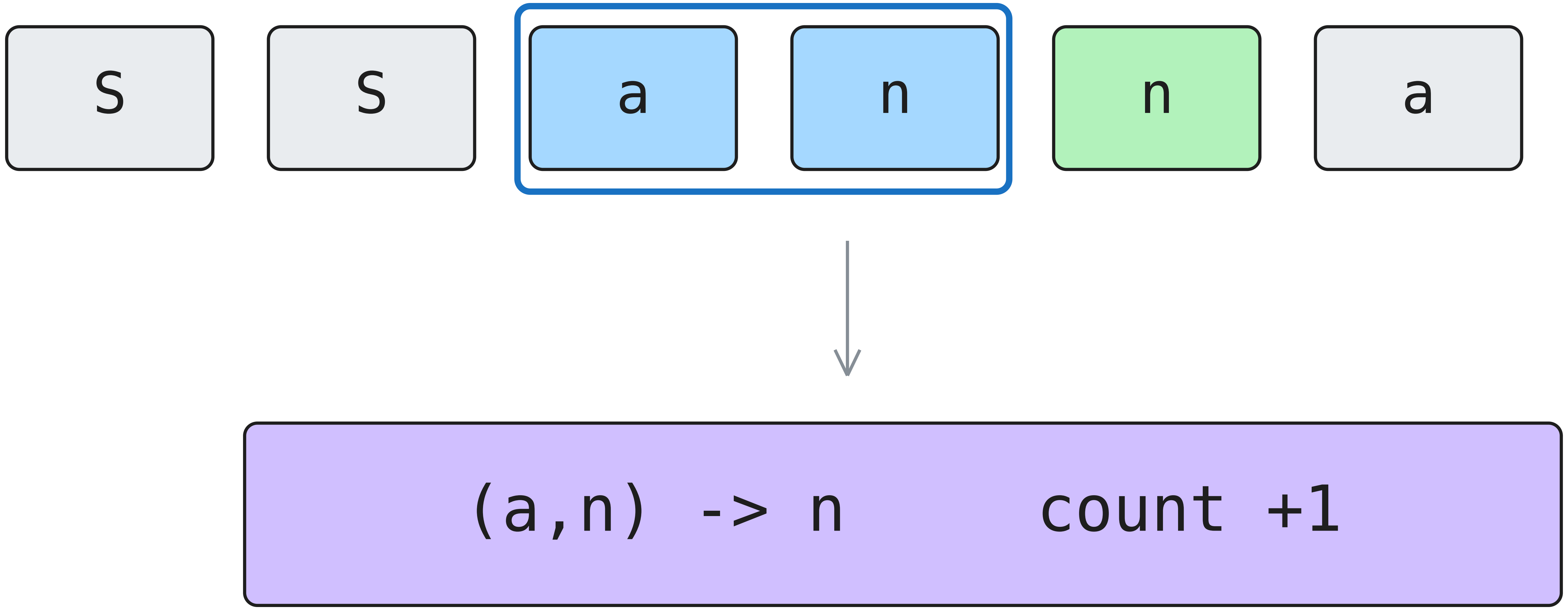
| | | | | | | |
|---|---|---|---|---|---|---|
| S | S | a | n | n | a | E |

```
(S,a) -> n      count +1
```

Move forward one position. Keep the last two tokens as context.

# Slide the window: anna, step 3 of 5.

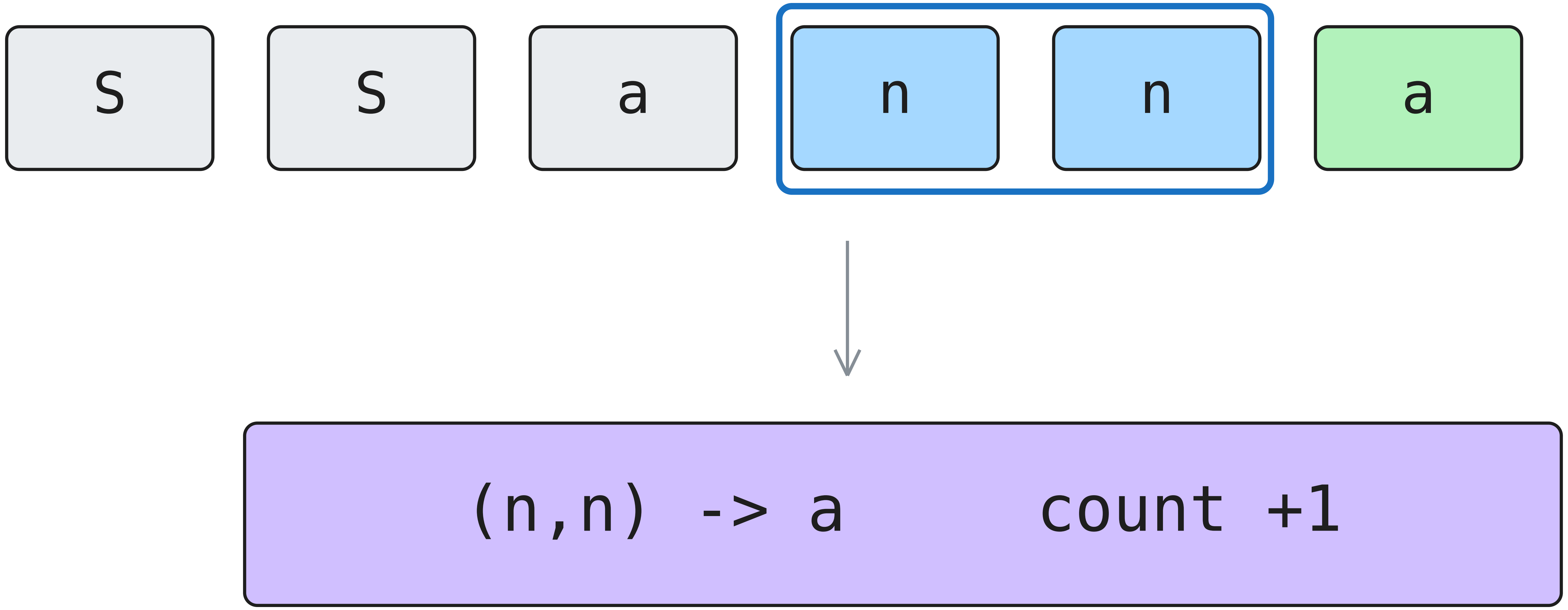
Blue cards are the context. The green card is the observed target.




Move forward one position. Keep the last two tokens as context.

# Slide the window: anna, step 4 of 5.

Blue cards are the context. The green card is the observed target.




Move forward one position. Keep the last two tokens as context.

# Slide the window: anna, step 5 of 5.

Blue cards are the context. The green card is the observed target.

S S a n n a E

```
(n,a) -> E      count +1
```

Stop after predicting END.

# Add ava. Combine identical observations.

Each context is a separate row. Columns are possible next tokens.

| | a | n | v | E |
|---|---|---|---|---|
| S,S | 2 | 0 | 0 | 0 |
| S,a | 0 | 1 | 1 | 0 |
| a,n | 0 | 1 | 0 | 0 |
| n,n | 1 | 0 | 0 | 0 |
| n,a | 0 | 0 | 0 | 1 |
| a,v | 1 | 0 | 0 | 0 |
| v,a | 0 | 0 | 0 | 1 |

ava adds these four:

```
(S,S) -> a
(S,a) -> v
(a,v) -> a
(v,a) -> E
```

7 observed context rows contain all 9 training observations.

# Divide each row by its own total.

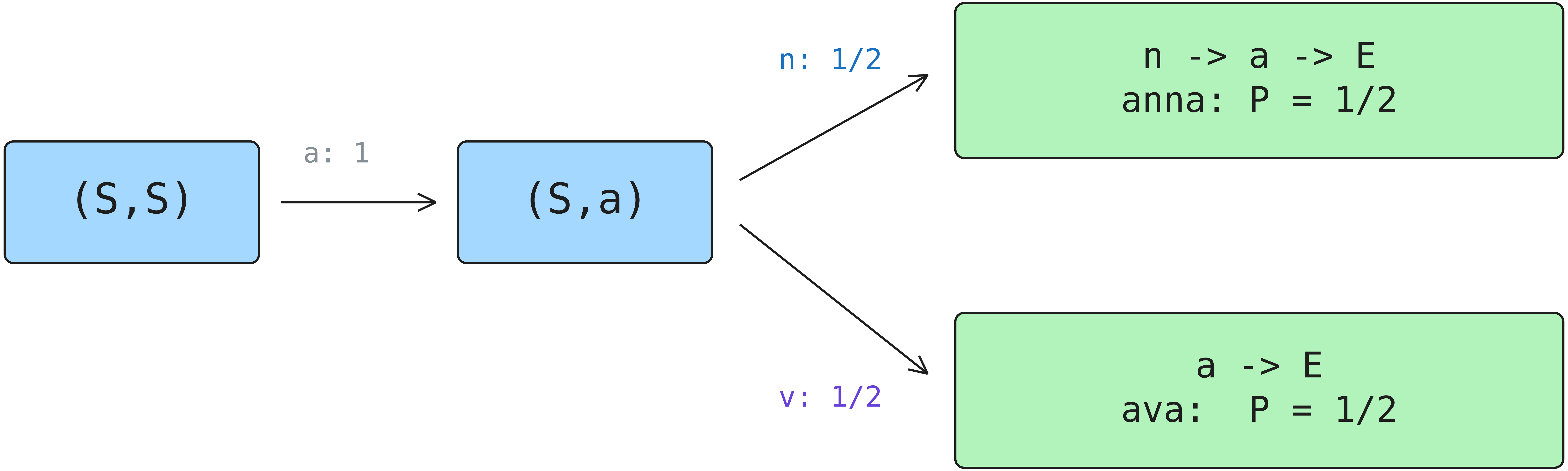
The denominator is the evidence for this context, not for the whole dataset.

| (S,a): n=1 v=1 | (a,n): n=1 |
|---|---|
| row total = 2 | row total = 1 |
| P(n \| S,a) = 1/2 | P(n \| a,n) = 1 |

Every observed row becomes its own probability distribution.

# Follow both possible generation paths.

Unsmoothed toy model. Every arrow after the branch has probability 1.




On this tiny corpus, the trigram generates only the two training names.

# Now ask it to score ana.

Evaluation follows the supplied name. It does not sample.

(S,S) -> a
P = 1

×

(S,a) -> n
P = 1/2

×

(a,n) -> a
P = 0

×

(n,a) -> E
P = 1

BIGRAM
P(ana) = 1/16

TRIGRAM
P(ana) = 0

A better training fit can reject a held-out spelling. Trigram NLL is infinite here.

# An empty cell is not an unseen row.

All these characters are known. What is missing is training evidence.

| | a | n | v | E |
|---|---|---|---|---|
| a,n | 0 | 1 | 0 | 0 |

(v,n)
no observations at all

$P(a \mid a,n) = 0/1 = 0$

0/0: row unspecified

The context was seen.
This target was not.

Counting supplies no
preferred distribution.

Zero in a known row is different from having no distribution for the row.

# Smoothing gives every outcome a ticket.

Add k=1. There are four outcomes: a, n, v, E.

## OBSERVED (a,n)

| a | n | n | v | E |
|---|---|---|---|---|

1 real ticket + 4 added tickets

`P(a | a,n) = 1/5`

## UNSEEN (v,n)

| a | n | v | E |
|---|---|---|---|

0 real tickets + 4 added tickets

`P(a | v,n) = 1/4`

Yellow = pseudo-counts, not extra training observations.

# Recompute the whole path after smoothing.

Every row changes, including rows that already gave positive probabilities.

| (S,S) -> a<br>3/6 | × | (S,a) -> n<br>2/6 | × | (a,n) -> a<br>1/5 | × | (n,a) -> E<br>2/5 |
|---|---|---|---|---|---|---|

P(ana) = 1/75

NLL = ln(75) / 4 ≈ 1.079372

Finite now. The denominator remains four predictions, including END.

# 100% can rest on one observation.

Original training names: anna and ava. No smoothing here.

```
n -> n        n -> a
```

Bigram: 2 matching observations

P(a | n) = 1/2

```
(a,n) -> n : once
```

```
(n,n) -> a : once
```

P(a | n,n) = 1

A count estimate of 100% describes the sample. It is not a universal rule.

# What if we add a third context token?

Separate constructed corpus: anna, enna, inna, onna. 20 predictions in total.

| 2 context characters | | 3 context characters |
| --- | --- | --- |
| (n,n) -> a<br>4 observations | → | (a,n,n) -> a : 1<br>(e,n,n) -> a : 1<br>(i,n,n) -> a : 1<br>(o,n,n) -> a : 1 |

Same target probabilities here. Each longer-context row now has less support.

# A longer query can have no matching row.

Same separate corpus: anna, enna, inna, onna. Query a supplied history ending nnn.

| n | n | n |
|---|---|---|

Predict the next token after this history.

TRIGRAM uses (n,n)
a followed it 4 times

4-GRAM uses (n,n,n)
0 observations

Known letters can form an unseen context. The unsmoothed row is unspecified.

# Each extra position multiplies possibilities.

Counting illustration only: alphabet a, n, v. Ignore boundaries for this frame.

| | | |
|---|---|---|
| a | n | v |

1 position: 3

| | | |
|---|---|---|
| aa | an | av |
| na | nn | nv |
| va | vn | vv |

2 positions: $3 \times 3 = 9$

3 positions: $3 \times 3 \times 3 = 27$

Each extra position multiplies the ordinary-context count by the alphabet size.

# Include the valid START contexts.

Capacity calculation: A = 26 letters. Two context positions; 27 next-token outcomes.

| (S,S) | (S,letter) | (letter,letter) |
|---|---|---|
| 1 | 26 | 676 |

$1 + 26 + 26^2 = 703$ possible rows

703 rows × 27 outcomes = 18,981 cells

Do not count (letter,S): START cannot appear after a letter.

# Possible rows multiply. Data does not.

Calculated capacities for 26 letters; these are not experiment results.

| | |
|---|---|
| 1 context character | 27 rows |
| 2 context characters | 703 rows |
| 3 context characters | 18,279 rows |
| 4 context characters | 475,255 rows |
| 5 context characters | 12,356,631 rows |

Original toy:

(4+1)+(3+1)

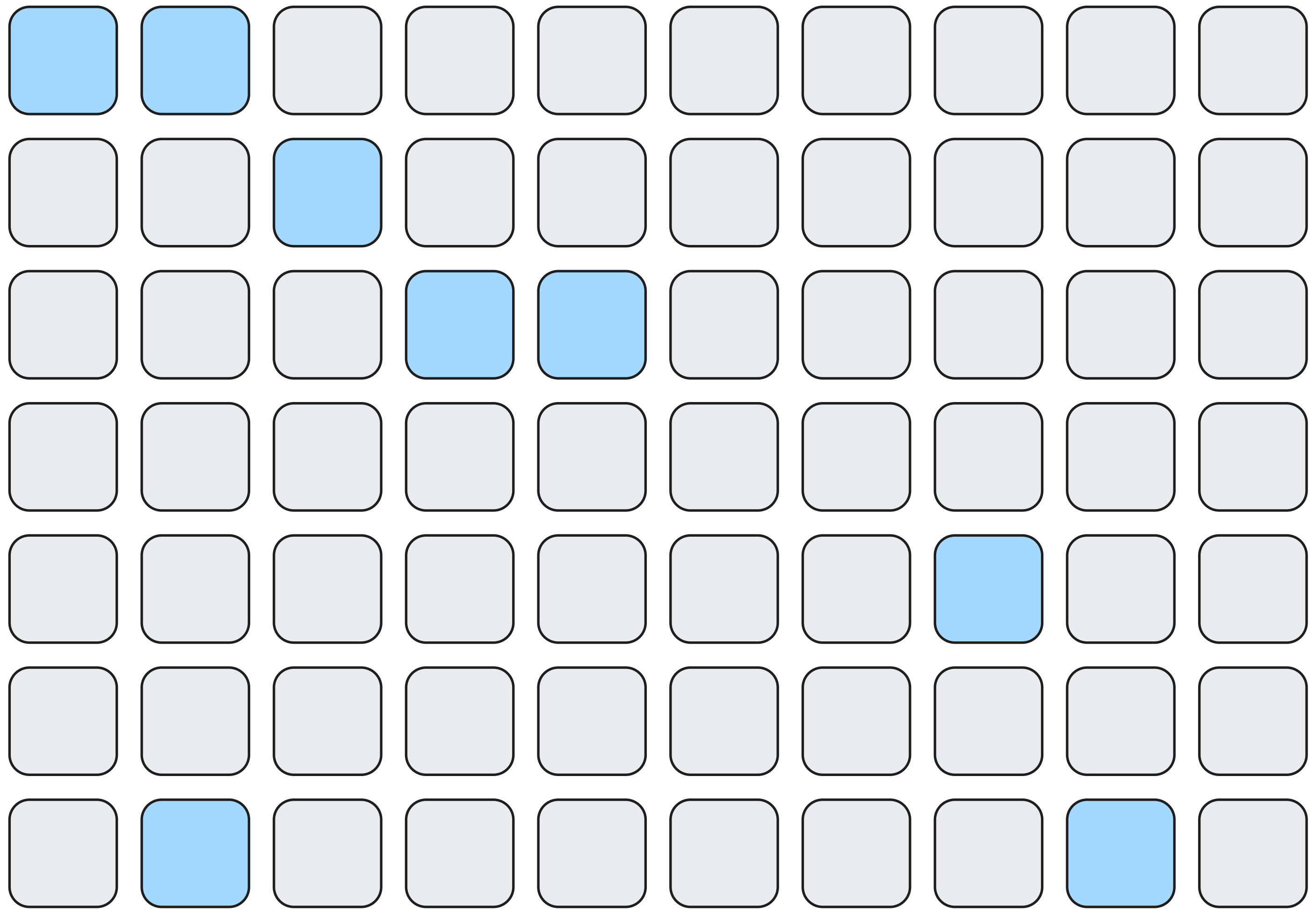
= 9

predictions at every context size

More possible groups for the same evidence; not every group will be observed.

# Sparse storage saves space, not evidence.

Illustrative grid. Store only counts that occurred during training.



Store only the blue cells.

**STORAGE**
Skip empty entries.

**EVIDENCE**
Missing observations
are still missing.

Smoothing can be calculated when needed; it does not add real observations.

# Three immediate responses, each with a limit.

The issue is context detail relative to available representative data.

### MORE DATA

More matching observations

Long contexts can still be unseen.

### SHORTER CONTEXT

Pool more evidence into each row

Some useful detail may be lost.

### SMOOTHING

Give known outcomes positive probability

An empty row still has no preference.

More data, less specific questions, and smoothing address different parts of the problem.

# Backoff: consult a shorter history.

Back to anna and ava. Illustration: fall back when the entire context is unseen.

(v,n)
no observations

use suffix →

n -> n : 1/2
n -> a : 1/2
v and END : 0

A simple fallback can reuse evidence we already have.

This rule only repairs absent rows. It does not repair every unseen continuation.

# Interpolation: let both histories contribute.

Original toy. Both component models use k=1; mixture weights are one-half.

TRIGRAM
P(a | a,n) = 1/5

BIGRAM
P(a | n) = 1/3

1/2 × 1/5 + 1/2 × 1/3 = 4/15

Mix defined probability rows with nonnegative weights that sum to one.

# Learn to share information across contexts.

Later in the series: shared representations. They still need data and good training.

**EPISODE 02**

Separate count rows
Useful context
Sparse evidence

→

**EPISODE 03**

Same bigram task
Weights + softmax
Learn probabilities

→

**EPISODE 07**

Embeddings + MLP
Shared parameters
Across contexts

A bigram weight table alone does not solve the longer-context sparsity problem.

# The tradeoff is context versus evidence.

A larger context can help when the data supports the distinctions it makes.

| | |
|---|---|
| MORE CONTEXT | Distinguish situations that a shorter history merges. |
| FIXED DATA | Each more specific row has at most the suffix row's evidence. |
| POSSIBLE IMPACT | Uncertain estimates, zeros, or entirely missing rows. |
| PRACTICAL RESPONSE | Gather evidence, pool it, smooth, or share information. |

The model can look certain while its estimate rests on very little evidence.

# Next: build it and test the tradeoff.

The coding companion uses the larger names dataset from Episode 01.

| | |
|---|---|
| BUILD | Counts, smoothing, evaluation, generation. |
| COMPARE | How do training and held-out loss change? |
| INSPECT | Which contexts are rare or entirely unseen? |

Will more context help on real data? We will measure it in the coding video.

# The foundations behind this episode.

Established n-gram methods; our own small examples and teaching sequence.


## Jurafsky & Martin · Speech and Language Processing

Chapter 3: N-gram Language Models / web.stanford.edu/~jurafsky/slp3

## Chen & Goodman · ACL 1996

An Empirical Study of Smoothing Techniques / aclanthology.org/P96-1041

## Bengio et al. · JMLR 2003

A Neural Probabilistic Language Model / jmlr.org/papers/v3/bengio03a.html


Tiny examples reveal the mechanism; held-out experiments measure practical performance.